

Microsoft Fabric

A Story Anyone Can Understand

From Data Lakes to Real-Time Intelligence — Explained Simply

DATA ENGINEERING

REAL-TIME INTELLIGENCE

POWER BI



It All Starts With Data

Every action, every second, generates data.



Gas Station Payment

Every swipe logs a transaction instantly



Doctor Visit Records

Diagnoses, prescriptions, test results — all stored



Grocery Store Scan

Inventory, pricing, and purchasing patterns captured



Every Tap, Click & Text

Your phone generates thousands of data points daily

! Where do you **put it all** — and what do you **do with it**?

Spreadsheets vs. Databases



Spreadsheet

- ✓ Great for small data
- ✓ Easy to use — anyone can start
- ✗ Breaks at scale — millions of rows
- ✗ One user at a time — no collaboration

SCALE CAPACITY

Low — thousands of rows



Database

- ✓ Handles millions of records with ease
- ✓ Multiple users — simultaneously
- ✓ Tables connected together — relational
- ✓ Fast search & query — structured language

SCALE CAPACITY

High — billions of records

vs

As businesses grew, **spreadsheets were no longer enough** — the world needed something far more powerful.

The Data Lake

When data got too big, messy, and fast for databases.



Massive Open Storage

A massive open storage space — dump anything in, any format, any size. Text, video, sensor logs, images — all welcome.



Data Flows In From Everywhere

Streams in continuously from sensors, apps, clicks, photos, GPS signals — like rivers feeding a lake, 24/7, never stopping.



The Problem: Data Swamp

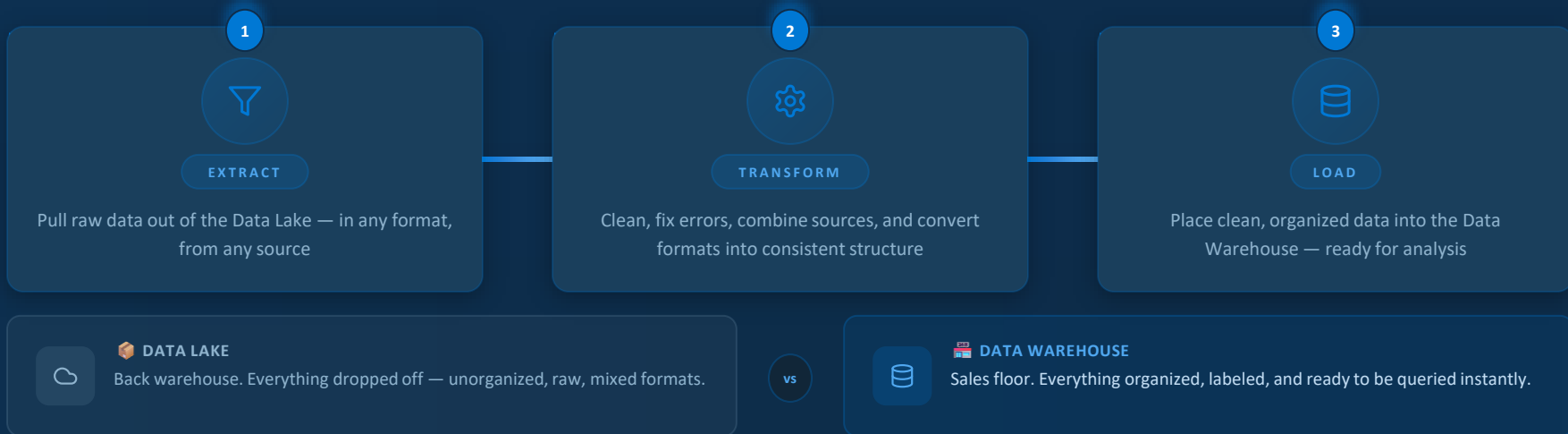
Without labels, structure, or organization, a Data Lake quickly becomes a Data Swamp — impossible to search or trust.



*A lake full of **unlabeled data** is just a swamp.*

The Data Warehouse & ETL

Taking raw data and turning it into something useful.



The Best of Both Worlds



Microsoft combined the Data Lake and Data Warehouse into one.



Data Lakehouse

Flexible storage + organized querying, in one system. No trade-offs — get the raw capacity of a lake with the structure of a warehouse.

✓ Flexible & Structured



OneLake

One shared storage location for the entire organization. No duplicates, no silos — every team works from the exact same copy of data.

✓ No Silos. No Duplicates.



One Source of Truth

Every team reads from the same data. Always consistent, always current — finance, ops, and engineering are finally on the same page.

✓ Consistent & Current



Microsoft Fabric is built on top of this foundation.



Moving Data Around: **Pipelines & Data Engineering**

Data does not move by itself — it needs a route.

1 PIPELINE



An automated route data travels along — like a water pipeline from reservoir to home.

3 DATA ENGINEERING



Designing, building, and maintaining all systems — the architects and plumbers of data.

2 DATA FACTORY



The tool that lets you build and manage pipelines — connect to hundreds of sources without complex code.

4 AUTOMATION



Runs on a schedule or triggers automatically when new data arrives. No manual work needed.



Data That Never Sleeps

LIVE

Real-Time Intelligence

Some data cannot wait hours — it needs to be acted on **in seconds**.



LIVE STREAMING

Eventstream

Live ticker of data streaming in **continuously** from any source — apps, sensors, devices, websites — all in real time.

 Ingesting...



CORE ENGINE

Real-Time Intelligence

Capture, process and **analyze data the instant it is created** — no waiting for batch jobs or overnight processing cycles.

• • • Millisecond latency



QUERY LANGUAGE

KQL

Kusto Query Language — ask questions of live data and get answers in seconds. Purpose-built for speed on streaming, time-series datasets.

`Bikes | where Station == "London" | summarize...`



STORAGE ENGINE

Eventhouse

High-speed database built **specifically for streaming, time-sensitive data** — ingests millions of events per second without breaking a sweat.

 Millions of events / sec



• REAL-WORLD EXAMPLE

- Know **exactly how many bikes are available** at every London station — **right now**. Not 10 minutes ago. Not from last night's batch. *This very second.*

NOW



Turning Data Into Pictures

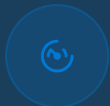
Most people don't want rows of numbers — they want *the story*.



Power BI

INSIDE MICROSOFT FABRIC

The visualization tool inside Microsoft Fabric. Turns raw data into **charts, graphs, maps** and tables.



Dashboard

AT-A-GLANCE VIEW

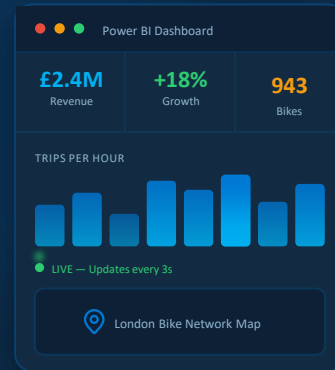
Like a car instrument panel — **speed, fuel, temperature** all visible at once. For business: revenue, inventory, alerts.



Live Updates

ALWAYS CURRENT

Connected to real-time data — the dashboard **updates automatically**. No manual refresh needed.



REAL-TIME DASHBOARD
auto-refreshing • no manual work

” The moment a bike is rented in London, **the dashboard reflects it**.

Teaching Computers to Think



HUMAN-LED ANALYSIS

Data Science

-  Analyze patterns in historical data
-  Build prediction models from that data
-  Answer: **What is likely to happen next?**
-  Example: What will sell most next Tuesday?

HISTORICAL PATTERN RECOGNITION



+



SELF-LEARNING SYSTEMS

Machine Learning / AI

-  Computer learns from past data **automatically**
-  Identifies patterns without being told what to look for
-  Gets **smarter over time** with more data
-  Runs inside Fabric — **no separate system needed**

SELF-IMPROVING NEURAL NETWORK



One Home for Everything

One login, one bill, one platform — six powerful workloads.



WORKLOAD 01

Data Engineering

Build pipelines and transform raw data into clean, structured insights.



WORKLOAD 02

Data Factory

Automate data workflows — connect hundreds of sources without complex code.



WORKLOAD 03

Real-Time Intelligence

Live streaming data and KQL queries — answers in seconds, not hours.



WORKLOAD 04

Data Science

AI and machine learning models trained on the same shared OneLake data.



WORKLOAD 05

Data Warehouse

Fast SQL queries at enterprise scale — organized, labeled, always ready.



WORKLOAD 06

Power BI

Live dashboards and reports for everyone — no data expertise required.



MICROSOFT FABRIC — IN SUMMARY

THE 30-SECOND VERSION

”

"Microsoft Fabric is like a complete data headquarters. It collects information from anywhere, keeps it all in one shared location, cleans and organizes it automatically, shows results on a live dashboard, and even predicts future trends — all in one place, one login, one bill from Microsoft."



SEE THE PROJECTS

github.com/svmalapaka

Explore code, notebooks & demos



CONNECT ON LINKEDIN

Sastry Malapaka

Let's talk data, Fabric & AI